TOWNSHIP INFRASTRUCTURE MANAGEMENT SYSTEM (TIMS)

SOFTWARE REQUIREMENTS SPECIFICATION

1. Preface

This document describes the main requirements of the Township Infrastructure Management System (TIMS) for users, developers, and testers.

2. Introduction

2.1 Purpose

TIMS helps a township manage infrastructure complaints, work orders, contractors, approvals, and payments.

2.2 Scope

The system allows residents/RWA representatives to report issues, staff to assign work, contractors to estimate and complete repairs, and the township to verify work and process payments.

2.3 Definitions

Term	Meaning
TIMS	Township Infrastructure Management System
RWA	Resident Welfare Association
Work Order	A request for repair or maintenance work
AMC	Annual Maintenance Contract
Estimate	Expected cost submitted by a contractor

3. System Users

- **RWA Representative** — reports issues and verifies completed work.
- **Desk Clerk** — reviews complaints, creates work orders, and assigns contractors.
- **Field Contractor** — inspects the site, submits estimates, and completes approved work.
- **Department Head** — approves or rejects every contractor estimate.
- **Finance** — manages contractors and AMC rates, checks invoices, and processes eligible payments.
- **Township COO** — oversees departments and views township work orders.

4. User Requirements

- The system shall allow RWA Representatives to submit complaints and track progress.
- The system shall allow Desk Clerks to review complaints and assign contractors.
- The system shall allow contractors to submit estimates and completion details.
- The system shall send every contractor estimate to the Department Head for approval.
- The system shall allow the RWA Representative to verify work or report a problem.
- The system shall allow Finance to check invoices and authorize payment after RWA verification.
- The system shall allow authorized users to view reports and work-order status.

5. Functional Requirements

5.1 Complaints and Work Orders

- The system shall give each complaint a unique reference number.
- The Desk Clerk shall review complaints and check for duplicates.
- A work order shall be created before contractor inspection.
- The Desk Clerk shall assign an eligible contractor to the work order.

5.2 Contractor and Estimate

- The contractor shall submit inspection details and an estimate using the applicable AMC rates.
- The system shall calculate the estimate total using the selected rates and quantities.
- Every estimate must be approved by the Department Head, regardless of its amount.
- The contractor shall start work only after approval.
- The contractor shall submit completion details and supporting evidence.

5.3 Verification and Payment

- The RWA Representative shall verify completed work or dispute it with a reason.
- Disputed work shall be corrected and submitted for verification again.
- Finance shall check the invoice against the approved estimate and completed work.
- Finance shall not authorize payment until RWA verification and financial checks are complete.
- The system shall record payment authorization and actual payment separately.

5.4 Administration

- The Department Head shall manage Desk Clerk accounts within their department.
- Finance shall manage contractor records, contract validity, and AMC rate cards.
- The COO shall oversee township departments and view work orders across them.
- The system shall restrict users to the data and functions permitted by their role.

6. Non-Functional Requirements

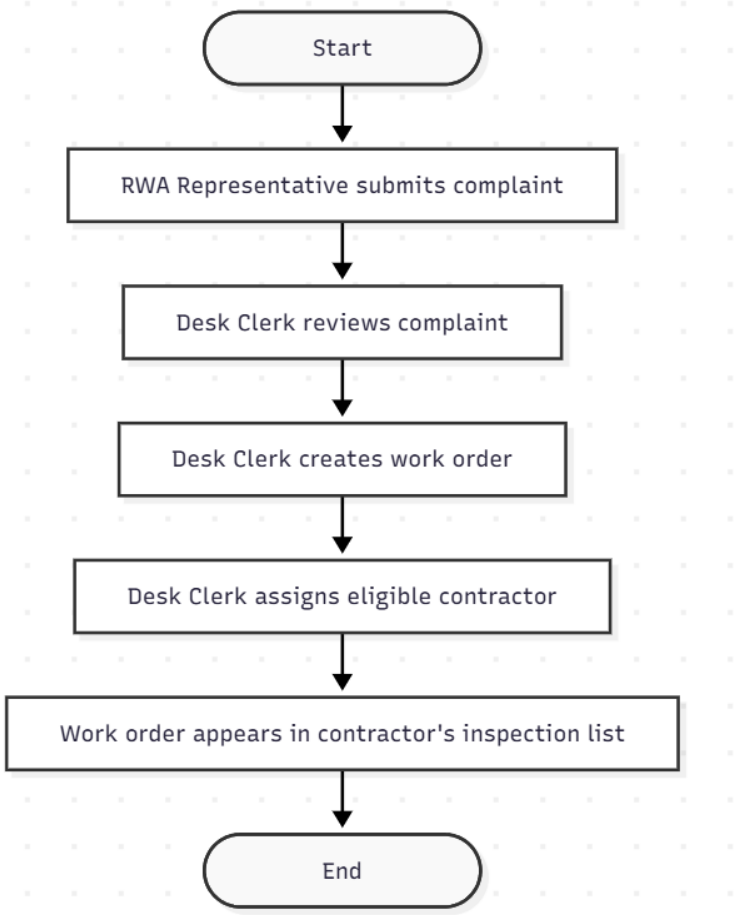
- **Performance:** Common actions should respond within 3 seconds; reports should generate within 5 seconds under agreed conditions.
- **Security:** Only authorized users shall access system functions.
- **Reliability:** The system shall maintain consistent work-order and payment records.
- **Usability:** The system shall have a simple interface that requires minimal training.

7. System Models

The complaint-to-payment workflow is modeled through four activity diagrams:

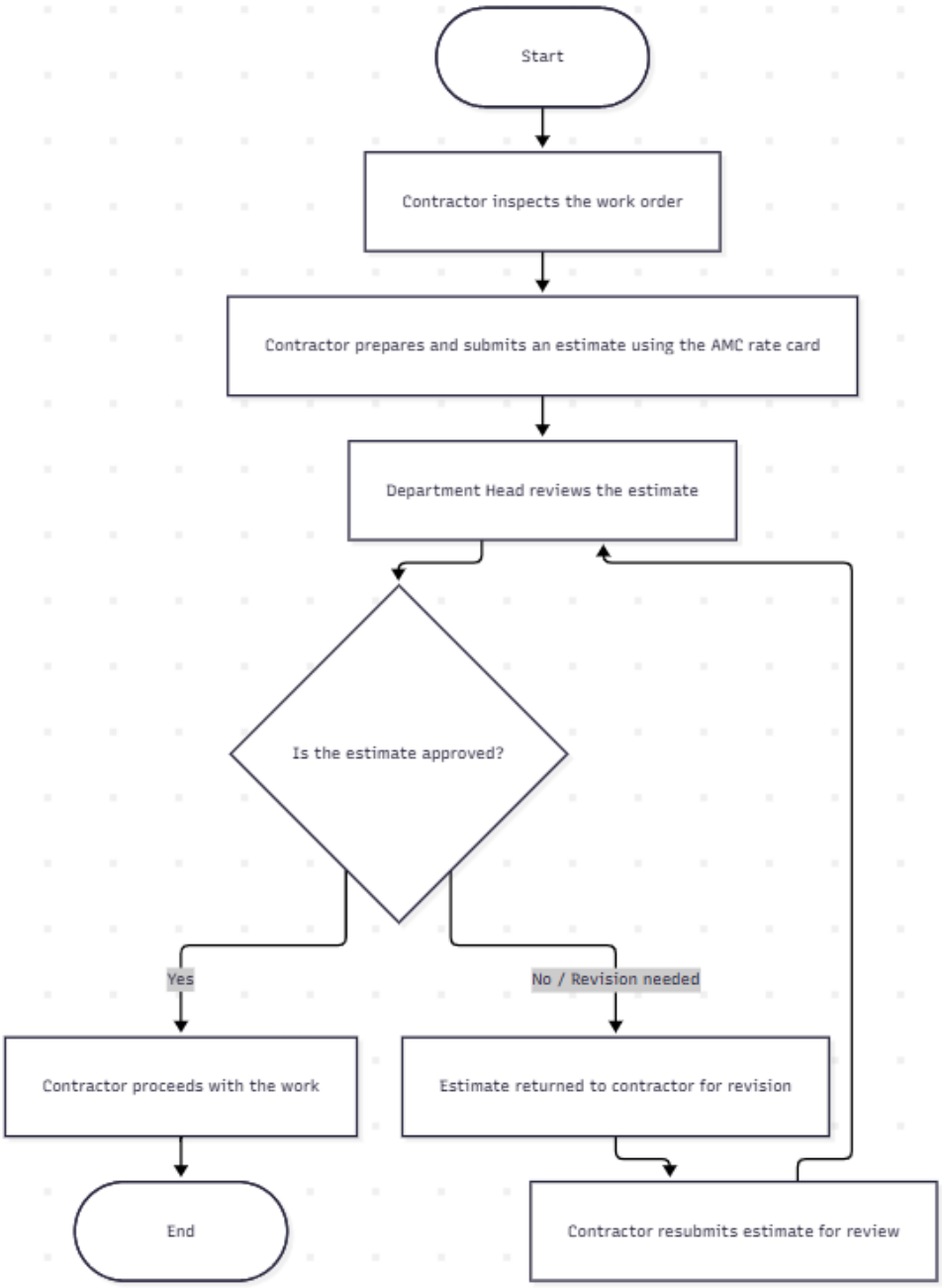
7.1 Activity Diagram 1: Complaint Submission and Triage

Captures the initial logging of an infrastructure issue by the RWA Representative and the Desk Clerk's triage steps to create a formal work order and assign a qualified contractor.



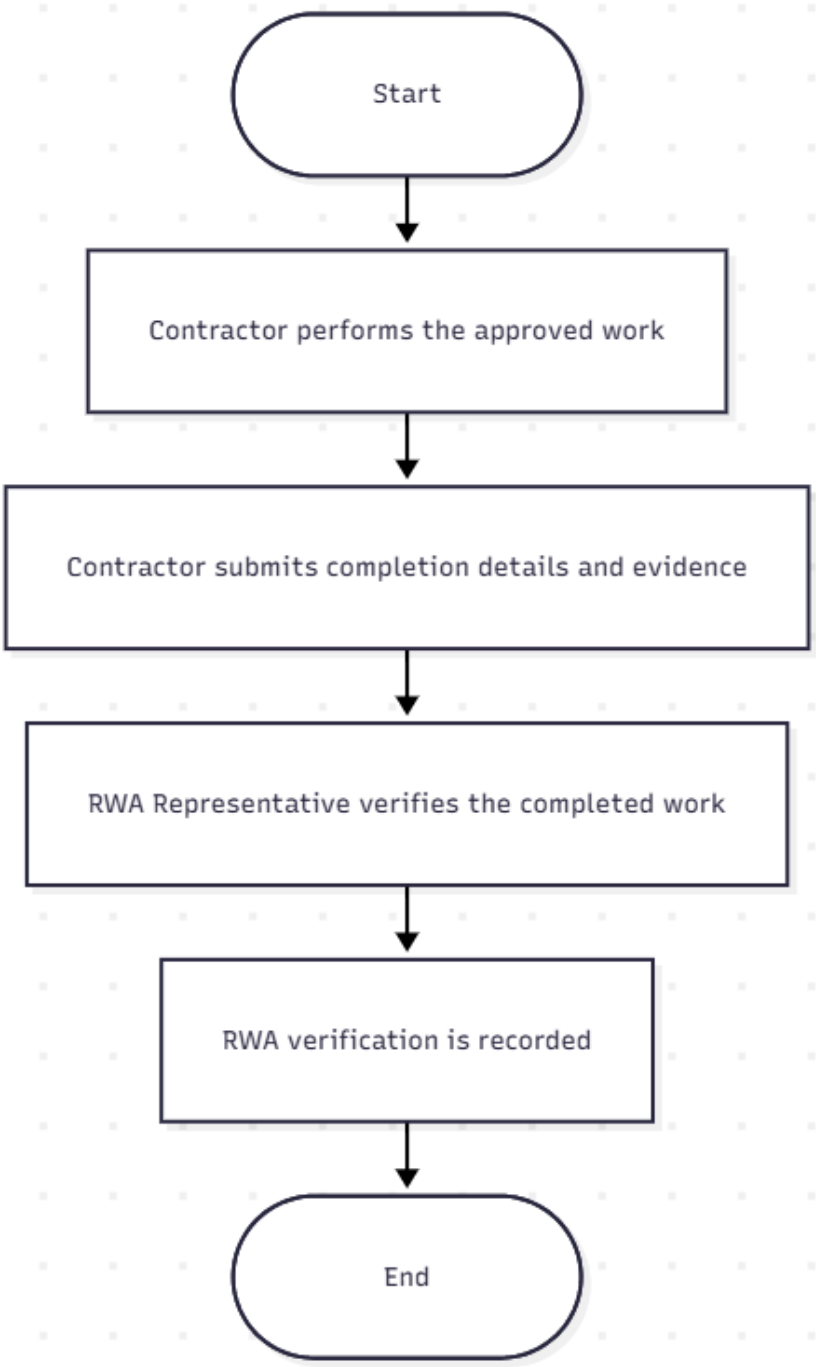
7.2 Activity Diagram 2: Inspection and Estimate Approval

Covers on-site inspection, cost estimation using AMC rate cards, and Department Head review with a revision loop for unapproved estimates.



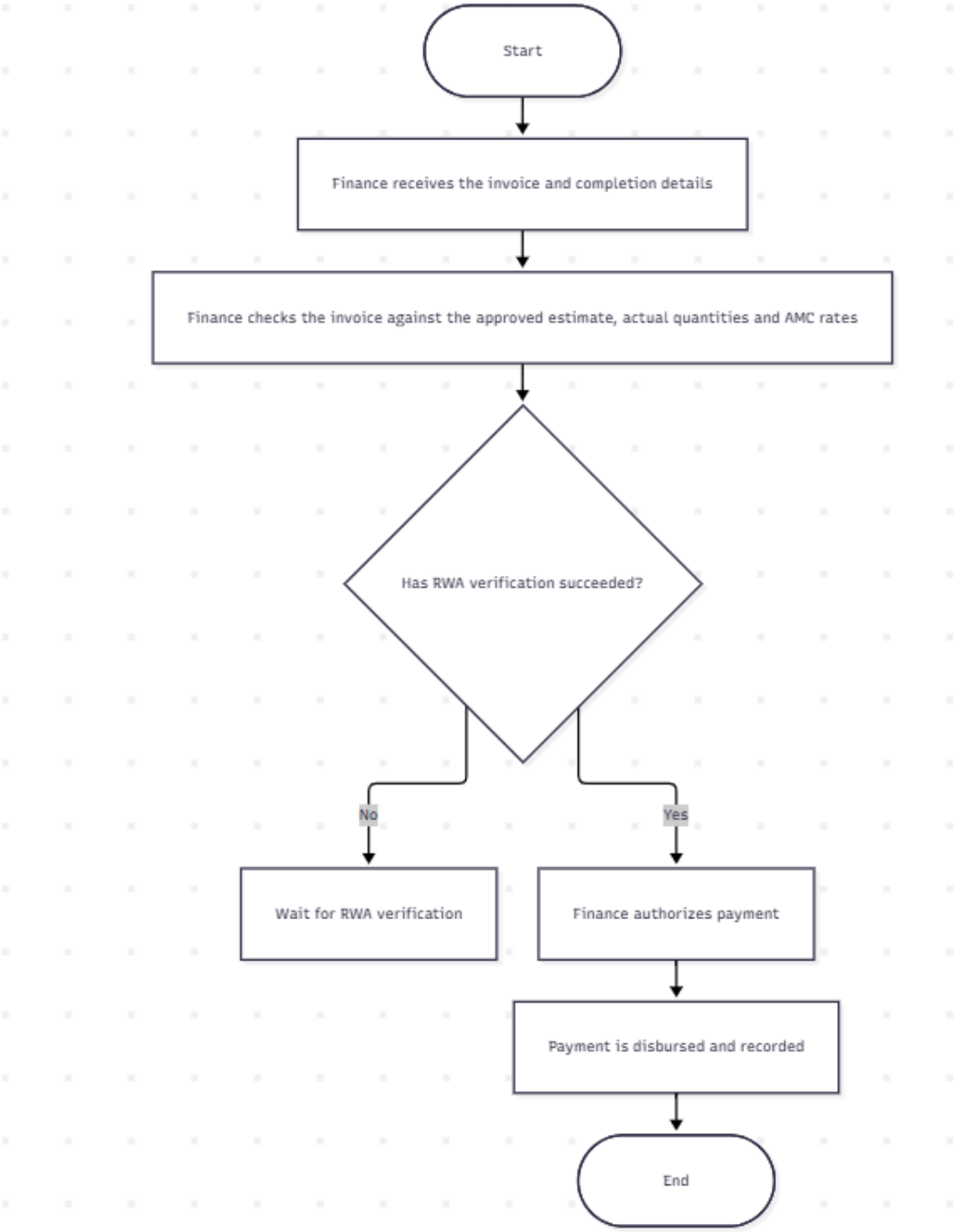
7.3 Activity Diagram 3: Work Execution and RWA Verification

Illustrates the repair phase where the contractor completes the approved work, attaches completion evidence, and the RWA representative validates and records the completion status.



7.4 Activity Diagram 4: Finance Audit and Payment

Details the financial reconciliation process: cross-checking contractor invoices against approved AMC rates and estimates, requiring RWA verification success, and disbursing payment.



8. Assumptions and Constraints

- Each township's data shall be kept separate from other townships.
- Every contractor estimate requires Department Head approval; there is no amount threshold.
- Payment cannot be authorized before RWA verification and Finance checks.

9. Conclusion

This document summarizes the requirements of TIMS, providing a direct baseline for designing and developing the township infrastructure management system.